

OFFICIAL TECHNICAL SPECIFICATION

System Design & Architecture Document (SDD)

SmartFarm Management Application

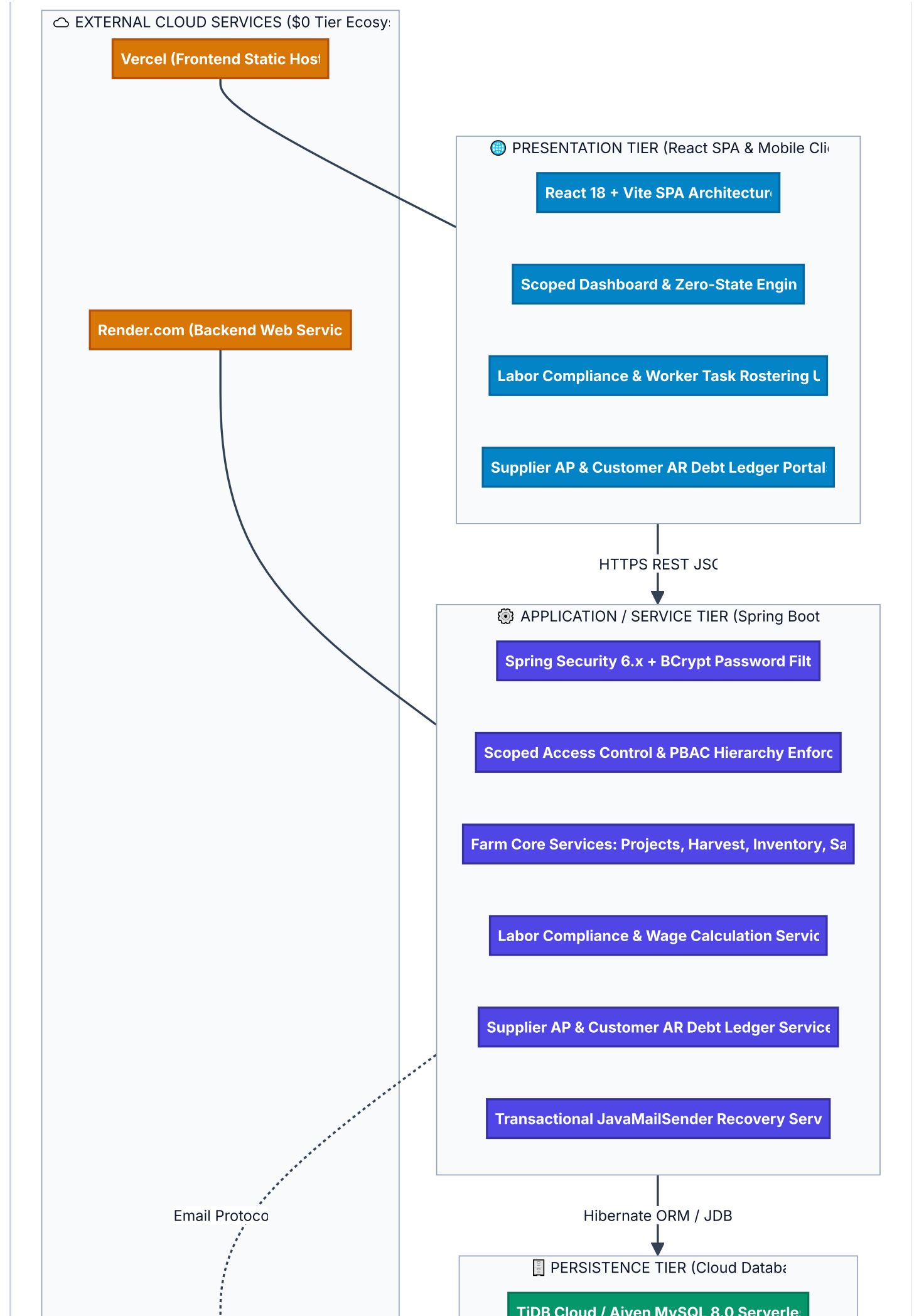
Version: 2.0 (Approved Baseline)

SDLC Stage: Phase 2 — System
Architecture, ERD & Module Design

Status: Approved Technical Blueprint

1. Architectural Overview & Technology Stack

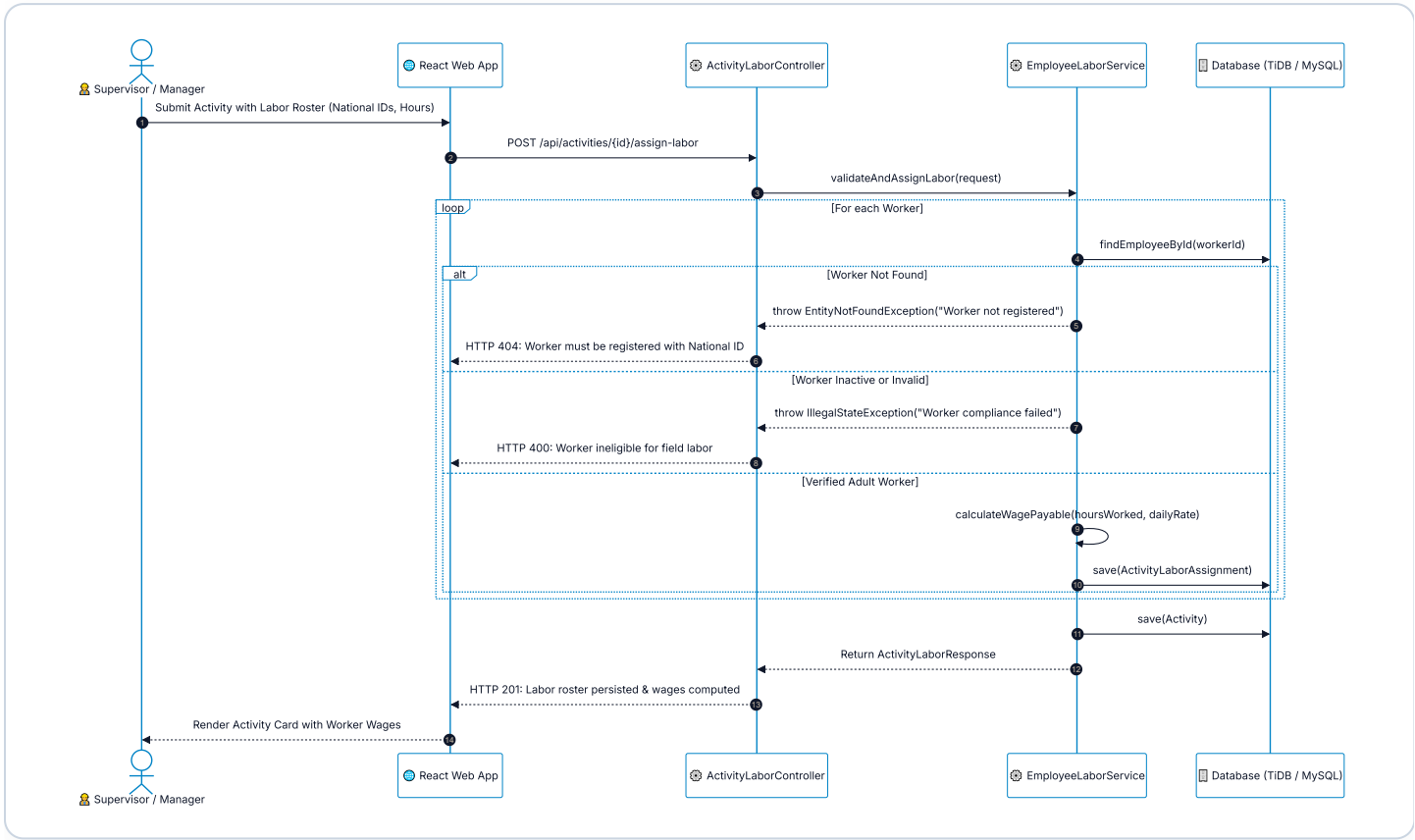
The SmartFarm system follows a **3-Tier Clean Architecture** separating presentation, business orchestration, and persistent storage:



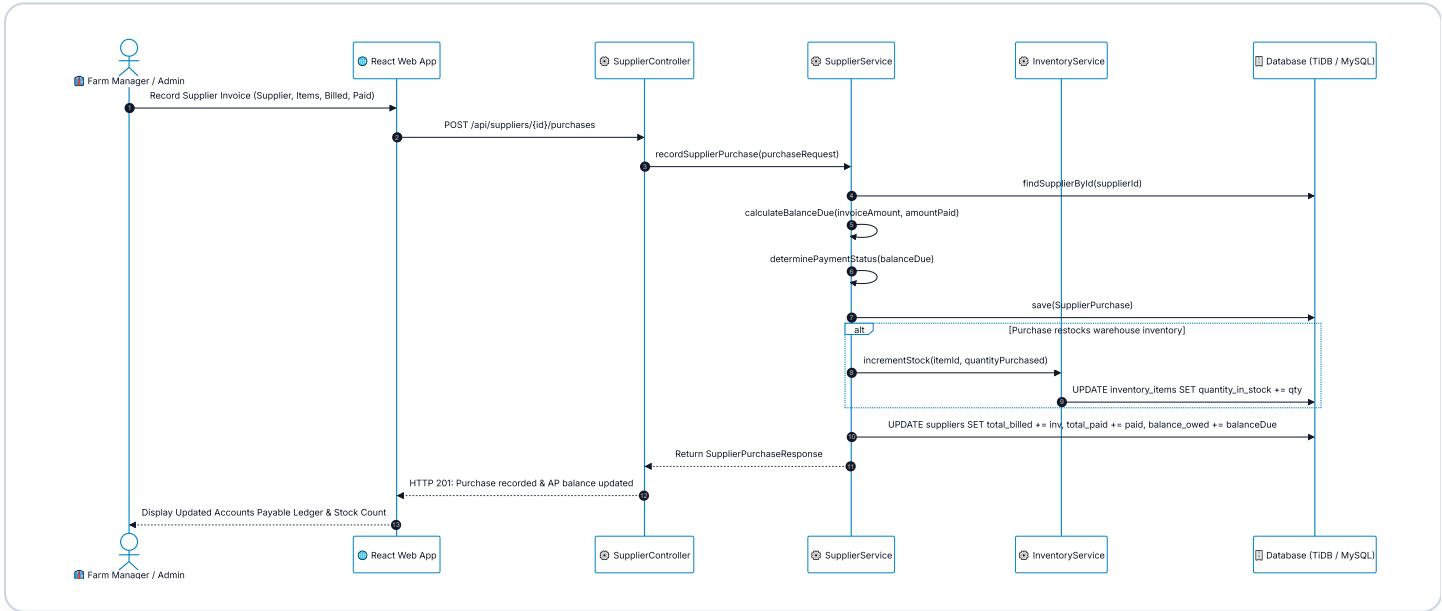
15 Relational Entities & Cascading Foreign Key

3. Sequence Diagrams for Core Business Workflows

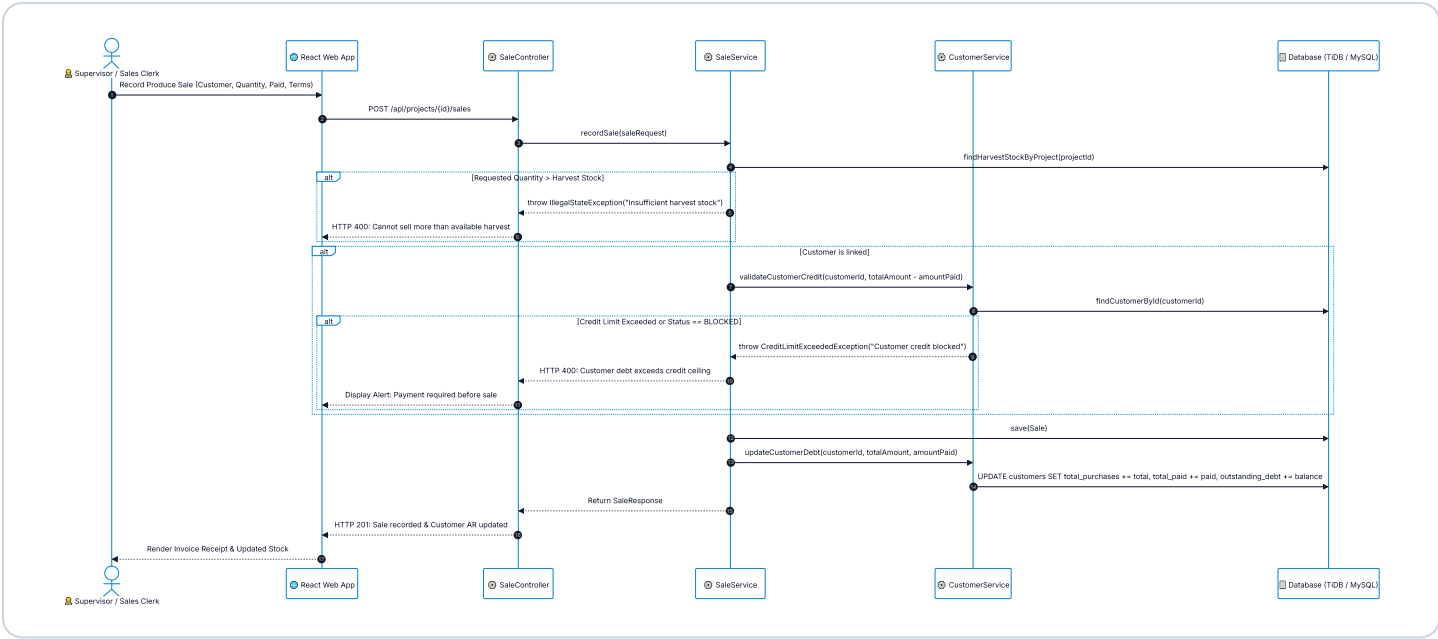
3.1 Adult Labor Compliance & Activity Assignment Sequence



3.2 Supplier Purchase Order & Stock Replenishment Sequence



3.3 Customer Credit Sale & Accounts Receivable Sequence



4. Complete REST API Specifications

4.1 Labor Compliance & Worker Endpoints

Endpoint	Method	Role Policy	Request Body	Description
`/api/employees`	`POST`	`ADMIN`, `MANAGER`	`CreateEmployeeRequest`	Registers verified adult worker with National ID.
`/api/employees`	`GET`	`ADMIN`, `MANAGER`, `SUPERVISOR`	*None*	Lists all active verified employees.
`/api/employees/{id}/status`	`PATCH`	`ADMIN`, `MANAGER`	`{ "status": "INACTIVE" }`	Toggles worker employment/activation status.
`/api/activities/{id}/labor`	`POST`	`ADMIN`, `MANAGER`, `SUPERVISOR`	`AssignLaborRequest`	Rosters workers for activity and calculates wages.
`/api/activities/{id}/labor`	`GET`	`ADMIN`, `MANAGER`, `SUPERVISOR`	*None*	Retrieves labor roster and wage details for a task.

4.2 Supplier & Accounts Payable Endpoints

Endpoint	Method	Role Policy	Request Body	Description
`/api/suppliers`	`POST`	`ADMIN`, `MANAGER`	`CreateSupplierRequest`	Registers external agricultural vendor.
`/api/suppliers`	`GET`	`ADMIN`, `MANAGER`	*None*	Lists suppliers with live Accounts Payable balances.
`/api/suppliers/{id}/purchases`	`POST`	`ADMIN`, `MANAGER`	`SupplierPurchaseRequest`	Logs purchase invoice and updates AP ledger.
`/api/suppliers/{id}/payments`	`POST`	`ADMIN`, `MANAGER`	`SupplierPaymentRequest`	Records debt settlement payment to supplier.

4.3 Customer & Accounts Receivable Endpoints

Endpoint	Method	Role Policy	Request Body	Description
`/api/customers`	`POST`	`ADMIN`, `MANAGER`, `SUPERVISOR`	`CreateCustomerRequest`	Registers produce buyer and assigns credit limit.
`/api/customers`	`GET`	`ADMIN`, `MANAGER`, `SUPERVISOR`	*None*	Lists customers with outstanding debt balances.
`/api/customers/{id}/payments`	`POST`	`ADMIN`, `MANAGER`, `SUPERVISOR`	`CustomerPaymentRequest`	Records debt remittance from customer.